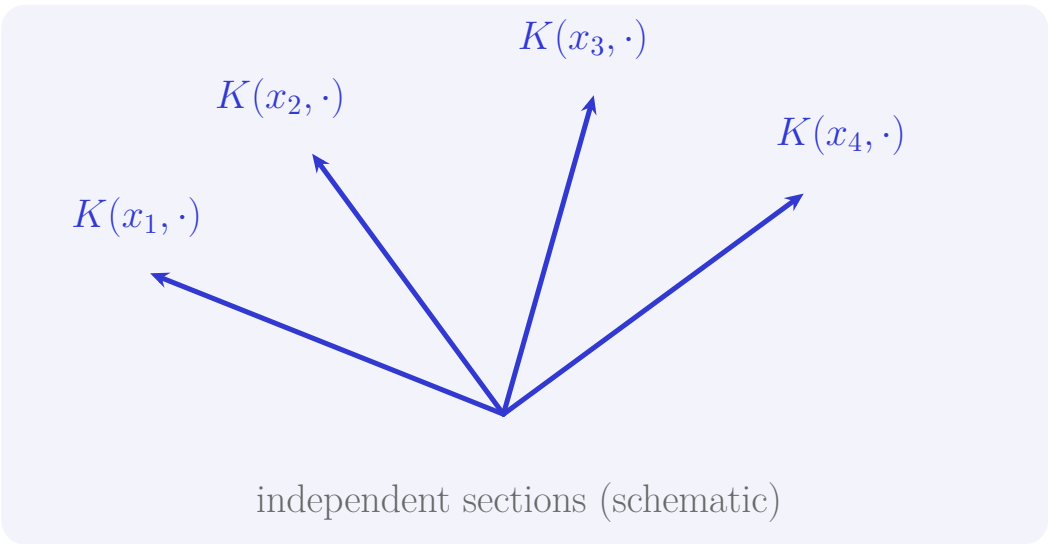


What changes in the NTK argument?

Universal-kernel guarantee

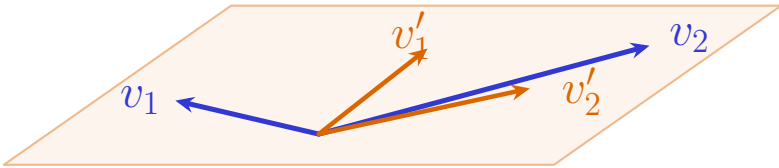


$$\sum_j a_j K(x_j, \cdot) = 0 \implies a_j = 0$$

distinct x_j

LoRA head at initialization

$$K^c(x, x') = \langle A_0 \phi(x), A_0 \phi(x') \rangle$$
$$\text{rank } K^c \leq r$$



$$V = [A_0 \phi(x_1) \cdots A_0 \phi(x_N)]$$

alternative bases; same free-coefficient fit

$$\Lambda V^\top = (\Lambda M^{-\top})(VM)^\top$$

$$T = 1 : \quad A_1 = A_0, \quad B_1 = -\eta D_0 (A_0 H)^\top$$

free coefficients; invertible M ; realizable candidate features required

Loo et al., Thm. 1: infinite width, unit sphere, nonzero training coefficients